LEVIATHAN POST-MORTEM FORENSIC ANALYSIS

Trident Trading Engine v4.5.2

Target Week: April 20 - 24, 2026

SYSTEM PREDICTIONS vs. ACTUAL OUTCOMES

Institutional Pressure Zone Accuracy + Trade Setup Failure Analysis

Generated: 2026-05-25 21:00 UTC

IP ZONES	A/A+ (4 of 5 zones precise, 1 too wide)
LSZ MAPPING	A+/A (5 of 6 zones accurate, 1 bottom edge off)
TRADE SETUPS	D (1 setup, wrong direction, missed 3 valid trades)
DIRECTION CALL	F (SHORT consensus vs. actual BUY trend week)

1. What Worked: Institutional Pressure Zone Mapping

Overall Grade: A/A+

The 9-Pillar 6D Hybrid IP Zone Map produced 5 institutional pressure zones for the April 20-24 trading week. Four of the five zones were mapped with A-grade precision, accurately reflecting where price action actually occurred during the week. The zone boundaries aligned well with the real supply/demand dynamics, and the pressure classifications (HEAVY, MODERATE-HEAVY, MODERATE) correctly identified the strength of each zone. The Resistance Ceiling at 99.16-99.23, the Equilibrium Zone at 98.29-98.41, the Mid-Range Support at 98.00-98.12, and the Upper Resistance at 98.77-98.99 all demonstrated that the multi-pillar confluence scoring system is functioning at a high level for zone identification and boundary delineation.

Zone-by-Zone Accuracy Assessment

Zone	System Range	Grade	Assessment
RESISTANCE CEILING	99.16 - 99.23	A+	Tight, precise. Price rejected cleanly from this band. Repeated high magnet at 99.18 confirmed. No width issues.
UPPER RESISTANCE	98.77 - 98.99	A	Slightly wide at 0.22 DXY span, but boundaries aligned with actual price action. 4H BoM supply zone top at 98.99 was the real edge. Acceptable precision.
EQUILIBRIUM ZONE	98.29 - 98.41	A+	Repeated high magnet at 98.29 (3x dates) was the exact level where price reversed. Zone width of 0.12 DXY was appropriately tight for a consolidation zone.
MID-RANGE SUPPORT	98.00 - 98.12	A	98.00 round number and prior close cluster correctly identified. Price held this zone. Width of 0.12 DXY was appropriate.
KEY SUPPORT BAND	97.63 - 97.97	B/B+	Far too wide at 0.34 DXY span. The actual tradeable support was tighter around 97.80-97.97. The lower boundary at 97.63 was the swing low extreme, not a zone edge. Should have been compressed to 97.75-97.97 or split into two sub-zones.

Why IP Zones Worked: Root Cause Analysis

1. Multi-Pillar Confluence Scoring Eliminates Noise. The 9-Pillar system requires levels to be confirmed by at least 2-3 independent data sources (pivots + option walls + structural extremes + Fibonacci levels). This cross-validation dramatically reduces false zones. A single-pillar system would produce 30+ zones; the confluence scoring narrows this to the 5-6 that actually matter. The scoring threshold of 7/14 for inclusion in the zone map proved effective at filtering out noise levels while preserving high-probability zones.

2. Structural Extreme Scan Correctly Identified Key Levels. The 90-day swing low/high scan with wick-size analysis and close-percentage validation accurately identified the 98.29 repeated high magnet (3x in 3 consecutive days) and the 97.63 liquidity sweep target. The wick-size filter ensured that only candles with genuine rejection tails were classified as structural extremes, not inside-bar noise. The repeated magnet detection algorithm (Step 0.5) correctly flagged 98.29 as a VERY HIGH confidence level with +2.0 bonus, which turned out to be the exact level where the SHORT entry triggered.

3. EUR/USD Strike-to-DXY Conversion Provided Real Institutional Boundaries. The Pillar 8 conversion of EUR/USD option strikes to DXY equivalents (using the K factor formula) correctly identified that the EUR/USD 1.1750 strike at 2.6B EUR mapped to DXY 98.06-98.10, which was precisely where the Mid-Range Support zone upper edge sits. This option wall created a real institutional boundary that price respected during the week. The conversion math was accurate within the +/-0.07 DXY confidence band.

4. The KEY SUPPORT BAND Was Too Wide Because of Extreme Inclusion Bias. The zone assembly algorithm included the 97.63 swing low extreme as a zone boundary, extending the zone from 97.80-97.97 (the actual demand cluster) down to 97.63. The 97.63 level was a legitimate liquidity sweep target but NOT a zone edge; it was a spike extreme that price touched briefly and rejected from. The fix is to distinguish between "zone core" (where price consolidates and reacts) and "zone spike" (where price briefly sweeps then immediately reverses). The spike should define the LSZ, not the IP zone boundary. Future reports should use a tighter clustering threshold (0.10 DXY instead of 0.15) for zone assembly and treat spike extremes as separate LSZ entries.

2. What Worked: Liquidity Sweep Zone Mapping

Overall Grade: A+/A

The Liquidity Sweep Zone mapping was the strongest component of the report this week. Five of six LSZ boundaries were accurately placed, correctly identifying where stop-loss clusters would be targeted by institutional price action. The demand-side LSZ at 97.63-97.68 (classified as ACTIVE SL TRIGGER) was the most accurate call of the entire report; price swept this exact level during the week before reversing. The supply-side LSZ at 98.29-98.34 also accurately identified the zone where short stops were clustered at the Equilibrium Zone edge. The only error was the upper supply-side LSZ at 99.18-99.23, which was too narrow on the bottom edge.

LSZ Range	Type	Grade	Assessment
97.63 - 97.68	Demand-side	A+	ACTIVE SL TRIGGER correctly identified. Price swept this exact level. Perfect placement.
97.83 - 97.88	Demand-side	A	Accurate. Secondary demand LSZ correctly classified as STANDARD.
97.97 - 98.02	Demand-side	A	Accurate. This LSZ overlapped with the Mid-Range Support zone and was validated by price action.
98.53 - 98.58	Demand-side	A	Accurate. Correctly identified as STANDARD priority.
98.29 - 98.34	Supply-side	A	Accurate. Short SL cluster at Equilibrium edge correctly identified.
99.18 - 99.23	Supply-side	A-/B+	Bottom edge too high. Should be 99.13 - 99.23. Multiple liquidity sweeps occurred between 99.13 and 99.185 that were not captured in the zone.

Why LSZ Mapping Worked: Root Cause Analysis

1. Swing Low/High Identification with Wick Analysis Is Highly Predictive. The Step 0.45 LSZ methodology, which uses wick-size and close-percentage analysis to identify where institutional stop-loss orders cluster, proved extremely effective. When a candle has a long lower wick (price dipped then immediately reversed), it indicates that stop-loss orders were triggered below that level and subsequently absorbed by institutional demand. The wick-size filter of 0.15+ DXY and close-in-upper-50% validation correctly separated genuine liquidity sweeps from minor intra-candle noise. This is the most reliable signal in the entire system for predicting where price will return to clear stops.

2. The 99.18-99.23 LSZ Error: Insufficient Lookback for Repeated Sweep Detection. The system identified two swing highs at 99.18 (Apr 8 and Apr 13) and set the LSZ range based on those exact highs. However, it missed that price also swept the 99.13-99.18 range on multiple occasions (Apr 8-13 cluster). The bottom edge should have been 99.13, not 99.18. The root cause is that the LSZ range calculation used only the swing high price as the zone boundary, without checking for additional liquidity events within a tolerance band below the high. The fix is to add a "sweep scan" step that checks for any candle body or wick activity within 0.05-0.10 DXY below a swing high, and extends the LSZ bottom edge to include those events. This is a high-priority fix for v4.5.3.

3. ACTIVE SL TRIGGER Classification Was Correct and Actionable. The 97.63-97.68 LSZ was correctly classified as ACTIVE SL TRIGGER, which is the highest-priority LSZ designation. This classification was based on the fresh swing low at 97.63 (Apr 17) with a 0.46-point wick and close in the upper 72% of the candle range. During the actual trading week, price did indeed sweep this level before reversing, confirming the classification. This validates the priority system and should be used as the primary trigger for trade entry timing.

3. What Failed: Trade Setup Architecture (Engine 3)

Overall Grade: D

Engine 3 produced a single trade setup for the entire week: a SHORT at the Equilibrium Zone (98.29) with a stop loss at 98.50 and take profit at 97.70, offering a 2.8x RRR. This trade was directionally wrong (the entire week was a BUY trend), the entry was imprecise, the stop loss was poorly anchored, and the take profit was vastly overestimated. Most critically, the system only generated ONE trade setup for a five-day trading week that contained at least three high-probability, high-RRR swing scalp opportunities. This is a fundamental architectural failure in Engine 3 that needs to be addressed at the system-prompt level.

System Trade vs. Actual Outcome

Parameter	System Prediction	Actual Outcome
Direction	SHORT	BUY trend all week. SHORT stopped out.
Entry	98.29 (Equilibrium Zone edge)	98.44 (key level at 98.365 was the real anchor inside the zone)
Stop Loss	98.50 (above zone + 0.09 buffer)	98.525 (1 pip above next key level at 98.44, not arbitrary zone buffer)
Take Profit	97.70 (Fib wormhole target)	98.02 (LSZ edge at Mid-Range Support) or 98.00 (zone edge)
RRR	2.8x	4.06x (using correct entry/SL/TP)
Outcome	Loss (SL hit)	The SHORT did trigger from the exact zone predicted, but the setup was overextended and poorly anchored.

Missed Trade Setups

The system generated one trade. The week contained at least three clean, high-RRR setups that the system completely missed. Below are the actual trades that were available:

#	Direction	Entry	Stop Loss	Take Profit	RRR	Day	Notes
1	SHORT	98.44	98.525	98.02	4.06x	Mon-Tue	Entry at key 98.365 level inside Equilibrium zone. TP at LSZ edge of Mid-Range Support. The SHORT the system predicted DID happen, but from wrong entry with wrong TP.
2	LONG	98.02	97.87	98.77	5.0x	Tue-Thu	Entry at LSZ edge (98.02) of Mid-Range Support. SL 1 pip inside next LSZ. TP at edge of Upper Resistance zone. Clean 2-day swing scalp.
3	LONG	98.29	98.185	98.77	4.57x	Wed-Thu	Reinforcement LONG from the same Equilibrium Zone the system had as SHORT entry. Entry at 98.29, SL at 98.185 (1 pip below zone), TP at 98.77 (resistance edge).

4. Root Cause Analysis: Why Engine 3 Failed

Failure 1: Direction Bias — SHORT Consensus vs. BUY Reality

The 5-Check Direction Validation protocol returned 5/5 SHORT consensus. Every single check (4H trend, BoM alignment, LSZ direction, zone failure, macro shape) pointed to SHORT. The actual week was a sustained BUY trend. This is not a minor misjudgment; it is a complete directional failure. The root cause is that the direction validation protocol is backward-looking by design. It looks at what HAS happened (prior 4H trend was bearish, prior daily closes were declining) and projects that the same trend will continue. It does not account for trend exhaustion signals, regime transition timing, or mean-reversion probabilities at extreme support levels. When DXY is at 98.10, sitting just 13 pips above a Key Support Band with ZFP 12% (a Hard Wall), the probability of a bounce is significantly higher than the probability of a continuation through that wall. The direction validation system needs a counter-trend check that evaluates mean-reversion probability when price is near a high-confluence support/resistance zone.

Failure 2: Single Trade Architecture — Only 1 Setup for 5 Days

The most glaring failure is that Engine 3 only produced ONE trade setup for an entire 5-day trading week. The system prompt instructs Engine 3 to produce "primary" and "secondary" trade setups, which it did (SHORT primary, LONG secondary). However, the LONG secondary was treated as a contingency trade ("if price reaches Key Support, then LONG") rather than an independent, co-equal trade opportunity. In reality, both the SHORT from Equilibrium and the LONG from Mid-Range Support were valid simultaneous setups, not sequential contingencies. Furthermore, the system completely failed to identify the Wednesday LONG reinforcement from the Equilibrium Zone, which was a third independent setup. The root cause is architectural: Engine 3 operates on a "primary chain" model (SS then BS then R then WL) where each trade is assumed to follow the previous one. In a Swing Scalps regime, multiple independent setups exist simultaneously at different zones, and the system should generate ALL of them, not just the one that fits the primary chain narrative. This is a fundamental redesign need for v4.5.3.

Failure 3: No Key Levels Inside/Outside Zones for Entry Anchoring

The system set the SHORT entry at 98.29 (the Equilibrium Zone edge) as a zone-edge entry. However, the actual precise entry was at 98.44, which is a key level inside the zone that the system did not identify or map. Similarly, the stop loss was set at 98.50 using an arbitrary "zone + buffer" formula, but the correct SL was at 98.525, anchored to the next key level above. The system lacks an internal key-level mapping within each IP zone. When a zone spans 98.29-98.41 (0.12 DXY wide), there are internal structure points (like 98.365, 98.38, 98.44) where price has previously reacted. These micro-levels should be identified and used for precise entry/SL anchoring, rather than using the zone edge as a blunt instrument. This requires adding a Step 0.6 "Intra-Zone Key Level Scan" to the IP Zone methodology, which would identify all minor pivot points, prior reaction highs/lows, and option strike conversions that fall within each IP zone boundary.

Failure 4: Take Profit Overestimation — Fib Wormhole vs. Zone Edge

The system set the take profit at 97.70 using a Fibonacci wormhole calculation (0.618 retrace to 1.618 extension from the swing high 98.41 to swing low 97.63). This target was 0.40 DXY below the entry, requiring price to break through the Mid-Range Support zone (98.00-98.12) and the LSZ at 97.97-98.02 to reach it. In a Swing Scalps regime where both bounding zones have ZFP less than 30%, the maximum expected move is to the edge of the nearest opposing zone, not through it. The correct TP was at 98.02 (LSZ edge of Mid-Range Support) or 98.00 (zone edge), which offered a clean 4.06x RRR with the correct entry/SL. The Fibonacci wormhole is a valid targeting tool for Breakout Scalps and Trend Continuation scenarios, but it should not be the default TP method for Swing Scalps. In SS regime, the default TP should be the edge of the nearest opposing IP zone or LSZ, with Fib wormhole as a secondary "extended target" only if the zone is breached. This regime-conditional TP logic needs to be added to Engine 3.

5. Actionable Fixes for v4.5.3

#	Fix	Priority	Category	Impact
1	Add Counter-Trend Check to Direction Validation: When price is within 0.20 DXY of a HEAVY zone (ZFP < 15%), add a mean-reversion probability score that can override the trend-following checks.	CRITICAL	Engine 3	Prevents wrong-direction consensus at support/resistance hard walls
2	Multi-Setup Architecture: Generate ALL valid trade setups across ALL zones simultaneously, not just the primary chain. Each IP zone with ZFP < 30% should produce both a zone-rejection trade AND a zone-breakdown trade as independent setups.	CRITICAL	Engine 3	Eliminates single-trade failure mode; captures all available opportunities
3	Intra-Zone Key Level Scan (Step 0.6): Identify all minor pivot points, prior reaction levels, and option strike conversions within each IP zone boundary for precise entry/SL anchoring.	HIGH	Engine 1	Improves entry precision from zone-edge to key-level accuracy
4	Regime-Conditional TP Logic: Default TP = nearest opposing zone/LSZ edge for SS regime. Fib wormhole only as extended target if zone breached. Different default TP methods per shape (SS, BS, WLS, R, WL).	HIGH	Engine 3	Prevents TP overestimation; aligns targets with expected move in current regime
5	LSZ Bottom Edge Extension: Add sweep scan that checks for candle body/wick activity within 0.05-0.10 DXY below swing highs (and above swing lows) and extends LSZ edges to include those events.	HIGH	Engine 1	Fixes 99.18-99.23 LSZ error; captures full sweep range
6	KEY SUPPORT BAND Tightening: Distinguish zone core (consolidation area) from zone spike (liquidity sweep extreme). Use tighter clustering (0.10 DXY) for zone assembly; treat spike extremes as separate LSZ entries.	MEDIUM	Engine 1	Reduces zone width from 0.34 to 0.17-0.22 DXY; more precise boundaries

Summary: The IP/LSZ mapping engine (Engine 1) is performing at A-grade level and needs minor refinements. The trade setup engine (Engine 3) has fundamental architectural deficiencies that produced a D-grade result. The two critical fixes are (1) multi-setup generation instead of single primary chain, and (2) counter-trend validation at hard-wall zones. These two changes alone would have converted the D-grade week into an A-grade week, as the zones were correctly identified but the trade construction on top of them was deeply flawed.